

Dissertation dev log

Week 13:

Current Goals for Full Dissertation session:

- Voronoi clipping implemented
- An implementation in GLSL, for performance comparison
- Research question made more precise and naturalistic
- Workplan make clearer
- Gather data
- Analyse data
- Specification for computing artefact that integrates societal, user, and business needs, demonstrating impact, usability and values to stakeholders
- Design rational, usage guidance, maintainability and architecture overview are provided
- Abstract
- Results and analysis section
- Discussions and conclusion
- Statistics addendum
- Artefact testing
- Critical addendum

Completed:

- A way to interact with the fragmented mesh
- A working Voronoi fracture in 2D for demonstration
- Working Voronoi fracture in 3D
- Comprehensive background done, covering triangulation, fracturing methods used in games, Graphics API's, Performance of Graphics API's, Point sampling, and Voronoi Clipping.
- Clearly stated research question
- Clear, feasible hypothesis, that is well scoped.
- An introduction that provides clear context for the project, and explains the need for the research, including positive impacts and who benefits. It should also highlight the motivation for the project.
- Professional, ethical, sustainable, legal, and social review carried out.
- A plan for security and privacy concerns has been written, and data is well managed.
- A methodology for how the project is going to be researched, and how its going to be evaluated
- A working way of profiling the results of the prototype

- A Critical Review of current technology is completed, that highlights where the gap in research is, and the how this project fills it.
- Project is ethically robust
- Project plan with gantt chart
- Profiling results that are easily replicated and are well documented.
- Fracture implemented in both OpenGL and Vulkan
- Readme that highlights development roadmap

Bugs

- Nothing currently, will be updated when bugs are found

Development notes:

Plane bisecting has been depreciated as the method of generating Voronoi diagrams, as it had a critical flaw that lead to the generation of malformed inner faces, that was difficult to solve. Instead, the Voronoi diagram is now generated by extracting it from a Delaunay triangulation generated using the Bowyer-Watson method. This has fixed the visual artifacting.

The development of the Voronoi clipping has been started, but significant difficulties have been found with the clipping to a complex mesh. The process of splitting intersecting triangles is difficult, and re-integrating them into a complete cell is challenging. If you re-triangulate split triangles, it can be rendered as is, but then you need to remove the cell, but keep the parts of the cell that are inside the mesh, to ensure its convex. This is difficult.

Images:

References:

Delaunay



Plane Bi-Secting

